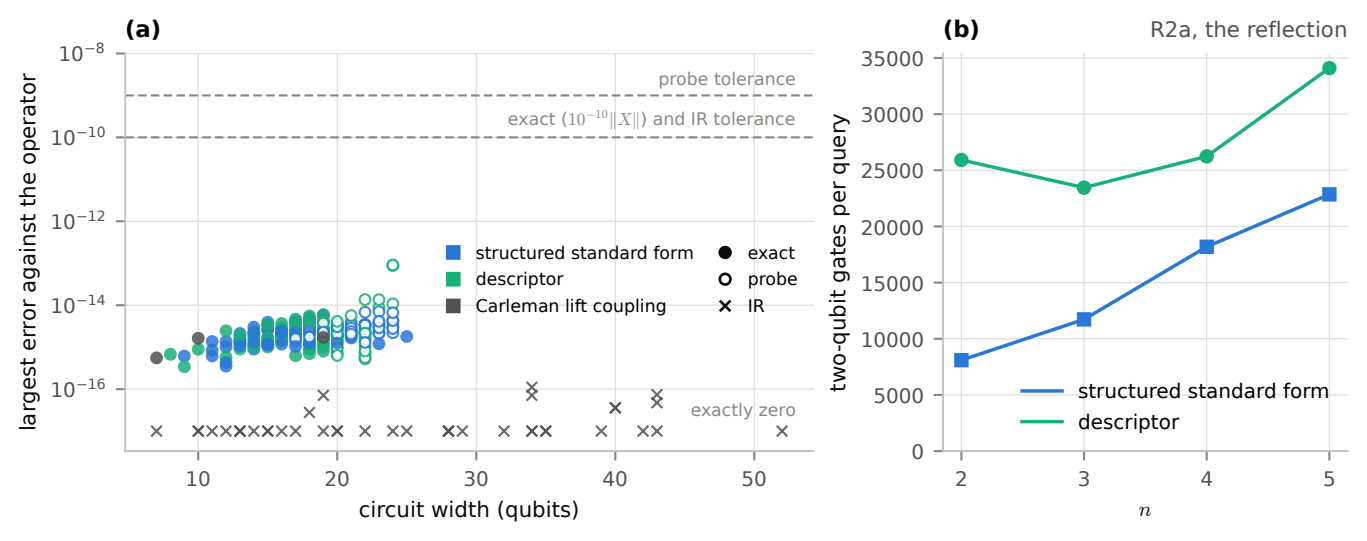

(a)
largest error against the operator
10^{-8}
10^{-10}
10^{-12}
10^{-14}
10^{-16}
probe tolerance
exact ($10^{-10}\|X\|$) and IR tolerance
structured standard form
descriptor
Carleman lift coupling
exact
probe
IR
exactly zero
10
20
30
40
50
circuit width (qubits)
(b)
R2a, the reflection
two-qubit gates per query
35000
30000
25000
20000
15000
10000
5000
0
structured standard form
descriptor
2
3
4
5
n